

Part I

Recitation Questions

Problem 1

February 21, 2026

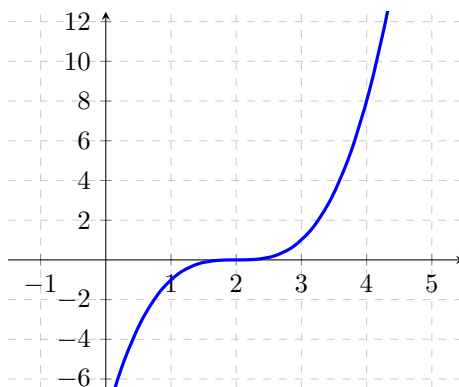
Problem 1 Determine whether the following statements are true or false and give either an explanation or a counterexample.

- (a) The function $f(x) = \sqrt{x}$ has a local maximum on the interval $[0, 1]$.

Explanation. False. Since $\sqrt{x}$ is increasing over the entire region $[0, 1]$, the only candidate for a local maximum would be $x = 1$. But by definition, endpoints are never local extrema. So $f(x) = \sqrt{x}$ has no local maximum on the interval $[0, 1]$.

- (b) If $f'(2) = 0$, then $x = 2$ is either a local maximum or local minimum of f .

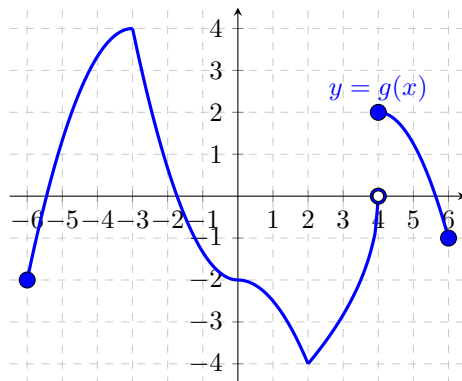
Explanation. False. Consider $f(x) = (x - 2)^3$. This has derivative $f'(x) = 3(x - 2)^2$, so that $f'(2) = 0$. We can see from the graph of f below, that $x = 2$ is not a local extremum for f .



Problem 2

February 21, 2026

Problem 2 The entire graph of a function g is given below.



Based on the graph of g , answer the questions below.

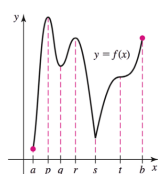
- (a) List the x -coordinates of all critical points of g .
Explanation. $x = -3$, $x = 0$, $x = 2$, and $x = 4$.
- (b) List the x -coordinates of all critical points of g where $g'(x) = 0$.
Explanation. $x = 0$
- (c) List the x -coordinates of all critical points of g where $g'(x)$ is **undefined**.
Explanation. $x = -3$, $x = 2$, and $x = 4$.
- (d) List the x -coordinates of all local maximums of g .
Explanation. $x = -3$ and $x = 4$.
- (e) List the x -coordinates of all local minimums of g .
Explanation. $x = 2$
- (f) List all intervals where g is both decreasing AND concave down.
Explanation. $(0, 2)$ and $(4, 6)$.
- (g) List all intervals where g is both decreasing AND concave up.
Explanation. $(-3, 0)$
- (h) List all intervals where g is both increasing AND concave down.
Explanation. $(-6, -3)$
- (i) List all intervals where g is both increasing AND concave up.
Explanation. $(2, 4)$
- (j) List the x -coordinates of all inflection points of g .
Explanation. $x = -3$, $x = 0$, and $x = 2$.

Problem 3

February 21, 2026

Problem 3

Problem 3 For each point in the interval (a,b) and identified on the graph below, determine if the function f has a critical point, a local max or min at that point.



Explanation. (p) The function f has a critical point and a local maximum at $x = p$.

(q) The function f has a critical point and a local minimum at $x = q$.

(r) The function f has a critical point and a local maximum at $x = r$.

(s) The derivative of f does not exist at $x = s$ because the function f has a corner at $x = s$. The function has a critical point and a local minimum at $x = s$.

(t) The function f has a critical point but not a local maximum or minimum at $x = t$.

Problem 4

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Problem 4 Locate the critical points and use the second derivative test to determine whether they correspond to local maxima or local minima. **EXPLAIN.**

$f(x) = (x + c)^4$ where c is a positive constant

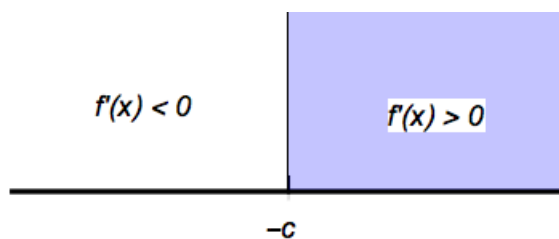
Explanation.

$$f'(x) = 4(x + c)^3, f''(x) = 12(x + c)^2$$

To find the critical points: $f'(x) = 0 = 4(x + c)^3$. This occurs when $x = -c$

Using the second derivative test: $f''(-c) = 12(-c + c)^2 = 0$. The second derivative test was inconclusive so we need to use the first derivative test.

Making a sign chart we see that if we plug something a little less than $-c$ into f' , we'll get a negative number. If we plug something a little greater than $-c$ into f' , we'll get a positive number.



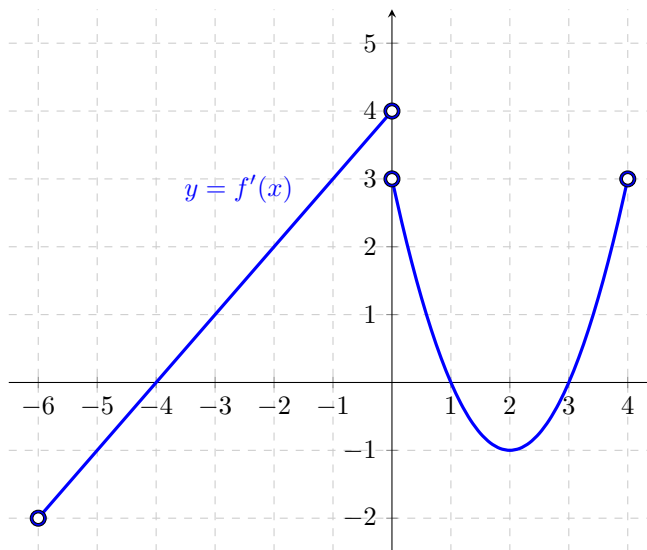
The derivative of f , f' , does not change sign on the intervals $(-\infty, -c)$ and on $(-c, \infty)$, therefore, we have the above chart. From the chart, we can conclude that f has a local minimum at $x = -c$, and that f also has a global minimum there. We could also have used our understanding of the family of functions $g(x) = x^4$ to reason $x = -c$ would be a minimum.

At $x = -c$, f has a local minimum by the first derivative test since $x = -c$ is a critical point of f and the sign of f' changes from negative to positive around $x = -c$.

Problem 5

February 21, 2026

Problem 5 Let f be a function that is continuous on its domain $(-6, 4)$. The graph of f' , the derivative of f , is given below.



Based on the graph of f' , answer the question below.

- (a) List x-coordinates of all critical points of f .

Explanation. Notice, $f'(-4) = f'(1) = f'(3) = 0$, and $f'(0)$ is not defined. Therefore, the function f has critical points at $x = -4$, $x = 0$, $x = 1$ and $x = 3$.

- (b) List x-coordinates of all critical points of f where $f'(x) = 0$.

Explanation. $x = -4$, $x = 1$ and $x = 3$.

- (c) List x-coordinates of all critical points of f where $f'(x)$ is **undefined**.

Explanation. $x = 0$

- (d) List x-coordinates of all local minimums of f .

Explanation. $x = -4$ and $x = 3$. At these points the sign of $f'(x)$ changes from negative to positive.

- (e) List x-coordinates of all local maximums of f .

Explanation. $x = 1$ At this point the sign of $f'(x)$ changes from positive to negative.

- (f) List all intervals where f is decreasing and concave down.

Explanation. $(1, 2)$ Notice, $f'(x)$ is negative and decreasing there.

- (g) List all intervals where f is decreasing and concave up.

Explanation. $(-6, -4)$ and $(2, 3)$. Notice, $f'(x)$ is negative and increasing there.

- (h) List all intervals where f is increasing and concave down.

Explanation. $(0, 1)$ Notice, $f'(x)$ is positive and decreasing there.

- (i) List all intervals where f is increasing and concave up.

Explanation. $(-4, 0)$ and $(3, 4)$. Notice, $f'(x)$ is positive and increasing there.

- (j) List x-coordinates of all inflection points of f .

Explanation. $x = 0$, and $x = 2$ $f'(x)$ is increasing on $(-6, 0)$ and decreasing on $(0, 2)$, then increasing on $(2, 4)$.

Problem 6

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Problem 6 Find the critical points of f on the given interval. Determine whether the function f has a local maximum, local minimum or no local extremum at each critical point. **EXPLAIN.**

(a) $f(x) = x\sqrt{2-x^2}$ on $(-\sqrt{2}, \sqrt{2})$.

Explanation.

$$\begin{aligned} f'(x) &= \sqrt{2-x^2} + x \left(\frac{1}{2\sqrt{2-x^2}}(-2x) \right) \\ &= \sqrt{2-x^2} - \frac{x^2}{\sqrt{2-x^2}} \\ &= \frac{2-x^2-x^2}{\sqrt{2-x^2}} \\ &= \frac{2(1-x^2)}{\sqrt{2-x^2}} \end{aligned}$$

Critical points of f occur where $f'(x) = 0$ or where $f'(x)$ does not exist. Solving $f'(x) = 0$ yields that $2(1-x^2) = 0$, or $x = \pm 1$. $f'(x)$ does not exist when $2-x^2 \leq 0$, but the points $x = \pm\sqrt{2}$ are not in the domain.

Let us use the Second Derivative Test to determine whether the function f has a local maximum or local minimum at the points where $x = \pm 1$.

$$f''(x) = \frac{d}{dx} \frac{2(1-x^2)}{\sqrt{2-x^2}} = \frac{2x(x^2-3)}{(2-x^2)^{\frac{3}{2}}}$$

Since $f''(-1) = 4$, the point $(-1, -1)$ is a local minimum by the Second Derivative Test. Since $f''(1) = -4$, the point $(1, 1)$ is a local maximum by the Second Derivative Test.

(b) $f(x) = x^3e^{-x}$ on $(-1, 5)$.

Explanation.

$$\begin{aligned} f'(x) &= 3x^2e^{-x} + x^3(-e^{-x}) \\ &= x^2e^{-x}(3-x) \end{aligned}$$

Notice that $f'(x)$ always exists, and so all of the critical points of f occur when $f'(x) = 0$. Solving this equation:

$$\begin{aligned} x^2e^{-x}(3-x) &= 0 \\ x^2(3-x) &= 0 \\ x = 0 \quad \text{or} \quad x &= 3 \end{aligned}$$

It is easier to use the First derivative test in order to determine whether the function f has a local maximum, local minimum or no local extremum at the point where $x = 0$. Since $-1 < 0 < 1$, $f'(-1) = 4e > 0$, and $f'(1) = 2e^{-1} > 0$, it follows that the sign of f' does not change at $x = 0$. The function f has **no local extremum** at $x = 0$.

Similarly, we will use the First derivative test in order to determine whether the function f has a local maximum, local minimum or no local extremum at the points where $x = 3$. Since $1 < 3 < 4$, $f'(1) = 2e^{-1} > 0$, and $f'(4) = -16e^{-4} < 0$ it follows that the sign of f' changes from positive to negative at $x = 3$. The function f has **a local maximum** at $x = 3$ by the First Derivative Test.

(c) $f(x) = x \ln \left(\frac{x}{5} \right)$ on $(0, 5)$.

Explanation.

$$\begin{aligned} f'(x) &= \ln \left(\frac{x}{5} \right) + x \cdot \frac{5}{x} \cdot \frac{1}{5} \\ &= \ln \left(\frac{x}{5} \right) + 1 \end{aligned}$$

Notice that $f'(x)$ exists for all values in $(0, 5)$, and so all of the critical points of f occur when $f'(x) = 0$. Solving this equation:

$$\begin{aligned} \ln \left(\frac{x}{5} \right) + 1 &= 0 \\ \ln \left(\frac{x}{5} \right) &= -1 \\ \frac{x}{5} &= e^{-1} \\ x &= 5e^{-1} = \frac{5}{e} \end{aligned}$$

Let's use the Second Derivative Test to determine whether the function f has a local maximum, local minimum or no local extremum at the points where $x = \frac{5}{e}$.

$$\begin{aligned} f''(x) &= \frac{d}{dx} \left(\ln \left(\frac{x}{5} \right) + 1 \right) \\ &= \frac{d}{dx} (\ln(x) - \ln(5) + 1) = \frac{1}{x} \end{aligned}$$

Since, $f''\left(\frac{5}{e}\right) > 0$, the point $\left(\frac{5}{e}, -\frac{5}{e}\right)$ is a local minimum by the Second Derivative Test.

Problem 7

February 21, 2026

Problem 7 Let $f(x) = \frac{1}{1+x^2}$. Find the following for f :

- (a) f' and f''

Explanation.

$$\begin{aligned} f'(x) &= \frac{(1+x^2)(0) - 1(2x)}{(1+x^2)^2} \\ &= \frac{-2x}{(1+x^2)^2} \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{(1+x^2)^2(-2) - (-2x)(2)(1+x^2)(2x)}{(1+x^2)^4} \\ &= \frac{-2(1+x^2) + 8x^2}{(1+x^2)^3} \\ &= \frac{6x^2 - 2}{(1+x^2)^3} \end{aligned}$$

- (b) *Critical points*

Explanation. Since $1+x^2 > 0$ for all x , f is differentiable over all real numbers. Thus all critical points of f occur when $f'(x) = 0$. But a fraction equals 0 if and only if its numerator equals 0. So

$$f'(x) = 0 \quad \implies \quad -2x = 0 \quad \implies \quad x = 0$$

Hence, the only critical point is $x = 0$.

- (c) *Local extrema (and check your answers with both the first and second derivative tests)*

Explanation. At $x = 0$, f' changes sign from positive to negative. Thus f goes from increasing to decreasing, and therefore by the first derivative test $x = 0$ is a local maximum of f .

For the second derivative test, we have that

$$f''(0) = \frac{6(0)^2 - 2}{(1+0^2)^3} = \frac{-2}{1} = -2 < 0$$

and thus we again conclude that $x = 0$ is a local maximum of f .

- (d) *Inflection points. EXPLAIN.*

Explanation. By the results in part

$$\begin{aligned}
 f''(x) &= \frac{6x^2 - 2}{(1 + x^2)^3} \\
 &= \frac{6(x^2 - \frac{1}{3})}{(1 + x^2)^3} \\
 &= \frac{6(x - \frac{1}{\sqrt{3}})(x + \frac{1}{\sqrt{3}})}{(1 + x^2)^3}
 \end{aligned}$$

which implies that the only candidates for inflection points are points where $x = \pm \frac{1}{\sqrt{3}}$. On the other hand, since $-1 < -\frac{1}{\sqrt{3}} < 0$, $f''(-1) = \frac{1}{2}$, and $f''(0) = -2$, it follows that f'' changes the sign at $x = -\frac{1}{\sqrt{3}}$. Similarly, since $0 < x = \frac{1}{\sqrt{3}} < 1$, $f''(0) = -2$, $f''(1) = \frac{1}{2}$, it follows that f'' changes the sign at $x = \frac{1}{\sqrt{3}}$, too. Since,

$$\begin{aligned}
 f\left(\frac{1}{\sqrt{3}}\right) &= \frac{3}{4} \\
 f\left(-\frac{1}{\sqrt{3}}\right) &= \frac{3}{4}
 \end{aligned}$$

The function f has inflection points at $\left(-\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$, $\left(\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$ because f is continuous at those points and f changes concavity around those points.

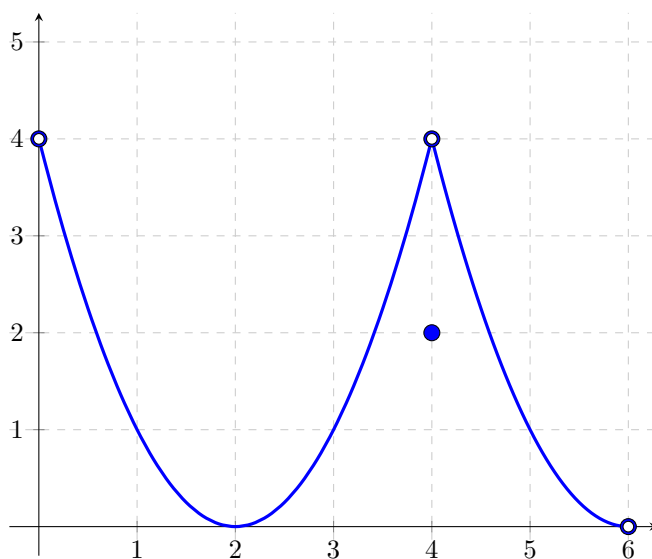
Problem 8

February 21, 2026

Problem 8 Sketch a possible graph of a function f that has the following properties:

- (a) f is defined on the interval $(0, 6)$.
- (b) f has no local maximums.
- (c) f has exactly two local minimums.

Explanation. This could be the graph of f . Notice two local minimums: at $x = 2$ and at $x = 4$.



Problem 9

February 21, 2026

Problem 9 Consider the parabola $f(x) = ax^2 + bx + c$ where a, b, c are constants. For what values of a, b, c is f concave up? For what values of a, b, c is f concave down?

Explanation.

$$\begin{aligned}f'(x) &= 2ax + b \\f''(x) &= 2a\end{aligned}$$

This means the sign of a will determine whether f is concave up or down. When $a < 0$, f is concave down, which makes sense because then the graph of f is a downward opening parabola. When $a > 0$, f is concave up, which makes sense because then the graph of f is an upward opening parabola. If $a = 0$, f has no concavity because it is a linear function.